

Missile_Reqs

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Requirement Set: Missile_Reqs

Description

Implementation Status

Total	Implemented	Justified	None
37	37	0	0

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
37	35	0	2	0	0

Change Information No change issue detected.

1 Cost

Requirement Type Functional

ID RFP - T

Description

The missile's unit cost shall not exceed \$500,000 (Fiscal Year 24-25).

Change Information No change issue detected.

Links

Artifact [top_level_arch.slx](#)



[Mass/Surge Production & Storage](#) (←Implemented by)

Artifact [UnitCost_Test.m](#)



[740105.521.1 in UnitCost_Test.m](#) (←Verified by)

Implementation Status

Total	Implemented	Justified	None
1	1	0	0

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	1	0	0	0	0

2 Production Run

Requirement Type Functional

ID RFP - T

Description

The missile shall have a production run of 1200 missiles per year for 10 years, as well as 100 more for developmental testing

Change Information No change issue detected.

Links

Artifact [top_level_arch.slx](#)



[Mass/Surge Production & Storage](#) (←Implemented by)

Artifact [Production Tests.m](#)



[740105.521.1 in Production Tests.m](#) (←Verified by)

Implementation Status

Total	Implemented	Justified	None
1	1	0	0

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	1	0	0	0	0

3 Surge Production

Requirement Type Functional

ID RFP - T

Description

The missile shall have a surge production capability of fabricating 24 units per week to 72 units per week, within a week, to be held indefinitely

Change Information No change issue detected.

Links

Artifact [top_level_arch.slx](#)

 [Mass/Surge Production & Storage](#) (↔Implemented by)

Artifact [Production_Tests.m](#)

 [740105.521.2 in Production_Tests.m](#) (↔Verified by)

Implementation Status

Total	Implemented	Justified	None
1	1	0	0

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	1	0	0	0	0

4 Range

Requirement Type Functional

ID RFP - T

Description

The missile's range shall be 500 nm from launch to impact

Change Information No change issue detected.

Links

Artifact [top_level_arch.slx](#)

 [Aerodynamics](#) (↔Implemented by)

Artifact [Cruise_Metrics_Test.m](#)

 [740105.524.5 in Cruise_Metrics_Test.m](#) (↔Verified by)

Implementation Status

Total	Implemented	Justified	None
1	1	0	0

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	0	0	1	0	0

5 Mark 41 VLS

Requirement Type Functional

ID RFP - T

Description

The missile shall be capable of being launched from land/
ship-based Mark 41 VLS

Change Information No change issue detected.

Links

Artifact [top_level_arch.slx](#)

 [Configuration](#) (↔Implemented by)

Artifact [Weight_Tests.m](#)

 [740021.521.3 in Weight_Tests.m](#) (↔Verified by)

Implementation Status

Total	Implemented	Justified	None
1	1	0	0

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	1	0	0	0	0

6 Warhead Size

Requirement Type Functional

ID RFP - T

Description

The warhead shall be sized to neutralize a single target

Change Information No change issue detected.

Links

Artifact [top_level_arch.slx](#)

 [Internal Configuration](#) (↔Implemented by)

Artifact [Warhead_Size_Destruction.m](#)

 [740039.521.1 in Warhead_Size_Destruction.m](#) (↔Verified by)

Implementation Status

Total	Implemented	Justified	None
1	1	0	0

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	1	0	0	0	0

7 Fixed Target Destruction

Requirement Type Functional

ID RFP - T

Description

The missile shall neutralize unhardened, fixed surface targets.

Change Information No change issue detected.

Links

Artifact [top_level_arch.slx](#)

 [Control System](#) (↔Implemented by)

Artifact [hit_the_target.m](#)

 [740105.521.1 in hit_the_target.m](#) (↔Verified by)

Implementation Status

Total	Implemented	Justified	None
1	1	0	0

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	1	0	0	0	0

8 Shelf Life

Requirement Type Functional

ID RFP - T

Description

The missile and every physical component shall have a shelf life of at least 10 years without maintenance.

Change Information No change issue detected.

Links

Artifact [top_level_arch.slx](#)

 [Mass/Surge Production & Storage](#) (↔Implemented by)

Artifact [Production Tests.m](#)

 [740105.521.3 in Production Tests.m](#) (↔Verified by)

Implementation Status

Total	Implemented	Justified	None
1	1	0	0

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	1	0	0	0	0

9 Entry into Service

Requirement Type Functional

ID RFP - T

Description

Design/Development shall commence no later than October 2026 and IOC shall occur no later than December 2031.

Change Information No change issue detected.

Links

Artifact [top_level_arch.slx](#)

 [Mass/Surge Production & Storage](#) (←Implemented by)

Artifact [Production Tests.m](#)

 [740105.521.4 in Production Tests.m](#) (←Verified by)

Implementation Status

Total	Implemented	Justified	None
1	1	0	0

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	1	0	0	0	0

10 Cruise Speed

Requirement Type Functional

ID RFP - O

Description

The missile should have a cruise speed of a minimum of Mach 3.

Change Information No change issue detected.

Links

Artifact [top_level_arch.slx](#)

 [Propulsion](#) (←Implemented by)

Artifact [Cruise Metrics Test.m](#)

 [740105.524.4 in Cruise Metrics Test.m](#) (←Verified by)

Implementation Status

Total	Implemented	Justified	None
1	1	0	0

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	1	0	0	0	0

11 Land Based Launch (HIMARS)

Requirement Type Functional

ID RFP - O

Description

The missile should be capable of being launched from land-based HIMARS and MLRS.

Change Information No change issue detected.

Links

Artifact [top_level_arch.slx](#)

☐ [Configuration](#) (↔Implemented by)

Artifact [Weight Tests.m](#)

 [740021.521.1 in Weight Tests.m](#) (↔Verified by)

Implementation Status

Total	Implemented	Justified	None
1	1	0	0

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	1	0	0	0	0

12 External Fighter

Requirement Type Functional

ID RFP - O

Description

The missile should be capable of being launched from an external fighter carriage.

Change Information No change issue detected.

Links

Artifact [top_level_arch.slx](#)

☐ [Configuration](#) (↔Implemented by)

Artifact [Weight Tests.m](#)

 [740021.521.1 in Weight Tests.m](#) (↔Verified by)

Implementation Status

Total	Implemented	Justified	None
1	1	0	0

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	1	0	0	0	0

13 Internal Fighter

Requirement Type Functional

ID RFP - O

Description

The missile should be capable of being launched from an internal fighter carriage.

Change Information No change issue detected.

Links

Artifact [top_level_arch.slx](#)

☐ [Configuration](#) (↔Implemented by)

Artifact [Weight Tests.m](#)

 [740021.521.1 in Weight Tests.m](#) (↔Verified by)

Implementation Status

Total	Implemented	Justified	None
1	1	0	0

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	1	0	0	0	0

14 Submarine

Requirement Type Functional

ID RFP - O

Description

The missile should be capable of being launched from a submarine VLS.

Change Information No change issue detected.

Links

Artifact [top_level_arch.slx](#)

 [Configuration](#) (↔Implemented by)

Artifact [Weight Tests.m](#)

 [740021.521.2 in Weight Tests.m](#) (↔Verified by)

Implementation Status

Total	Implemented	Justified	None
1	1	0	0

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	1	0	0	0	0

15 Air Launch Conditions

Requirement Type Functional

ID RFP - O

Description

The missile should be capable of air launch at Mach 0.85 from 30000 ft.

Change Information No change issue detected.

Links

Artifact [top_level_arch.slx](#)

 [Configuration](#) (↔Implemented by)

Artifact [Cruise Metrics Test.m](#)

 [740105.524.3 in Cruise Metrics Test.m](#) (↔Verified by)

Implementation Status

Total	Implemented	Justified	None
1	1	0	0

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	1	0	0	0	0

16 Dimensions

Requirement Type Functional**ID** RFP - O**Description**

The missile's maximum dimensions should be 15 x 15 x 144 in.

Change Information No change issue detected.**Links****Artifact** [top_level_arch.slx](#)
☐ [Configuration](#) (↔Implemented by)
Artifact [LaunchBox_Size_Test.m](#)
☐ [740021.521.1 in LaunchBox_Size_Test.m](#) (↔Verified by)
Implementation Status

Total	Implemented	Justified	None
1	1	0	0

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	1	0	0	0	0

17 Data Links

Requirement Type Functional**ID** RFP - O**Description**

The missile should contain Datalink commands to permit in-flight updates.

Change Information No change issue detected.**Links****Artifact** [top_level_arch.slx](#)
☐ [Avionics Package](#) (↔Implemented by)
Artifact [Avionics tests.m](#)
☐ [function Datalink\(testCase\)](#) (↔Verified by)
Implementation Status

Total	Implemented	Justified	None
1	1	0	0

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	1	0	0	0	0

18 Low-Level Ingress**Requirement Type** Functional**ID** RFP - O**Description**

The missile should fly < 500 ft AGL for the last 10% of its mission range.

Change Information No change issue detected.**Links****Artifact** [top_level_arch.slx](#)☐ [Approach](#) (←Implemented by)**Artifact** [Lowlevel ingress Test.m](#) [740105.524.1 in Lowlevel ingress Test.m](#) (←Verified by)**Implementation Status**

Total	Implemented	Justified	None
1	1	0	0

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	0	0	1	0	0

19 HDBT & Naval Target Destruction**Requirement Type** Functional**ID** RFP - O**Description**

The missile should neutralize HDBT and move naval targets.

Change Information No change issue detected.**Links****Artifact** [top_level_arch.slx](#)☐ [Control System](#) (←Implemented by)**Artifact** [hit the target.m](#) [740105.521.1 in hit the target.m](#) (←Verified by)**Implementation Status**

Total	Implemented	Justified	None
1	1	0	0

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	1	0	0	0	0

20 Factor Safety

Requirement Type Functional

ID MIL-M-8856B

Description

Factor of safety will be a minimum of 1.5 for in-flight, storage, and transportation.

Change Information No change issue detected.

Links

Artifact [top_level_arch.slx](#)

 [Structural Material](#) (←Implemented by)

Artifact [Material Tests.m](#)

 [740105.521.1 in Material Tests.m](#) (←Verified by)

Implementation Status

Total	Implemented	Justified	None
1	1	0	0

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	1	0	0	0	0

21 Failure under Ultimate Mission Load

Requirement Type Functional

ID MIL-M-8856B

Description

The structure will not fail under the ultimate load condition outlined in the mission profile

Change Information No change issue detected.

Links

Artifact [top_level_arch.slx](#)

 [Structural Material](#) (←Implemented by)

Artifact [Material Tests.m](#)

 [740105.521.2 in Material Tests.m](#) (←Verified by)

Implementation Status

Total	Implemented	Justified	None
1	1	0	0

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	1	0	0	0	0

22 Extreme Temp

Requirement Type Functional

ID MIL-STD-1522A

Description

Materials will be constructed based on extreme temperature cases seen during the mission profile

Change Information No change issue detected.

Links

Artifact [top_level_arch.slx](#)

 [Structural Material](#) (↔Implemented by)

Artifact [Material Tests.m](#)

 [740105.521.3 in Material Tests.m](#) (↔Verified by)

Implementation Status

Total	Implemented	Justified	None
1	1	0	0

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	1	0	0	0	0

23 Heating Effects

Requirement Type Functional

ID MIL-M-8856B

Description

Missile design shall provide for the cumulative heating effect from internal and external heating

Change Information No change issue detected.

Links

Artifact [top_level_arch.slx](#)

 [Thermal Material](#) (↔Implemented by)

Artifact [Material Tests.m](#)

 [740105.521.4 in Material Tests.m](#) (↔Verified by)

Implementation Status

Total	Implemented	Justified	None
1	1	0	0

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	1	0	0	0	0

24 Fatigue Life

Requirement Type Functional

ID MIL-M-8856B

Description

The fatigue life of the structure will equal or exceed the specified life

Change Information No change issue detected.

Links

Artifact [top_level_arch.slx](#)

 [Structural Material](#) (↔Implemented by)

Artifact [Material_Tests.m](#)

 [740105.521.5 in Material_Tests.m](#) (↔Verified by)

Implementation Status

Total	Implemented	Justified	None
1	1	0	0

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	1	0	0	0	0

25 Proof Pressure

Requirement Type Functional

ID SMC-S-006

Description

Missile shall sustain specified proof pressure without any detrimental deformation

Change Information No change issue detected.

Links

Artifact [top_level_arch.slx](#)

 [Structural Material](#) (↔Implemented by)

Artifact [Material_Tests.m](#)

 [740105.521.6 in Material_Tests.m](#) (↔Verified by)

Implementation Status

Total	Implemented	Justified	None
1	1	0	0

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	1	0	0	0	0

26 Minimum Proof Pressure

Requirement Type Functional

ID SMC-S-006

Description

Minimum proof pressure will be 1.1 time the maximum expected operating pressure

Change Information No change issue detected.

Links

Artifact [top_level_arch.slx](#)

 [Structural Material](#) (↔Implemented by)

Artifact [Material_Tests.m](#)

 [740105.521.7 in Material_Tests.m](#) (↔Verified by)

Implementation Status

Total	Implemented	Justified	None
1	1	0	0

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	1	0	0	0	0

27 Stiffness

Requirement Type Functional**ID** SMC-S-006**Description**

All components will have enough stiffness to handle deformation at limit load

Change Information No change issue detected.**Links****Artifact** [top_level_arch.slx](#) [Structural Material](#) (←Implemented by)**Artifact** [Material Tests.m](#) [740105.521.8 in Material Tests.m](#) (←Verified by)**Implementation Status**

Total	Implemented	Justified	None
1	1	0	0

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	1	0	0	0	0

28 Loading with Cracks

Requirement Type Functional**ID** SMC-S-006**Description**

Components will handle loads even with existing crack

Change Information No change issue detected.**Links****Artifact** [top_level_arch.slx](#) [Structural Material](#) (←Implemented by)**Artifact** [Material Tests.m](#) [740105.521.9 in Material Tests.m](#) (←Verified by)**Implementation Status**

Total	Implemented	Justified	None
1	1	0	0

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
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1	1	0	0	0	0
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29 Fracture Toughness

Requirement Type Functional

ID SMC-S-006

Description

Fracture toughness in materials must be at a minimum of 60% of the plain strain toughness

Change Information No change issue detected.

Links

Artifact [top_level_arch.slx](#)

☐ [Structural Material](#) (↔Implemented by)

Artifact [Material Tests.m](#)

 [740105.521.10 in Material Tests.m](#) (↔Verified by)

Implementation Status

Total	Implemented	Justified	None
1	1	0	0

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	1	0	0	0	0

30 Fabrication

Requirement Type Functional

ID SMC-S-006

Description

Fabrication of materials will be within acceptable variable ranges and perform reliably

Change Information No change issue detected.

Links

Artifact [top_level_arch.slx](#)

☐ [Structural Material](#) (↔Implemented by)

Artifact [Material Tests.m](#)

 [740105.521.11 in Material Tests.m](#) (↔Verified by)

Implementation Status

Total	Implemented	Justified	None
1	1	0	0

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	1	0	0	0	0

31 Internal Pressure

Requirement Type Functional

ID MIL-STD-1522A

Description

The system will maintain safe conditions for pressure level inside necessary for flight conditions outlined

Change Information No change issue detected.

Links

Artifact [top_level_arch.slx](#)

 [Structural Material](#) (↩Implemented by)

Artifact [Material Tests.m](#)

 [740105.521.12 in Material Tests.m](#) (↩Verified by)

Implementation Status

Total	Implemented	Justified	None
1	1	0	0

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	1	0	0	0	0

32 Ignition System Protection

Requirement Type Functional

ID MIL-STD-1901A

Description

For safe storage, the ignition system will not arm during exposure to electronic attacks(ie. EMR, ESD, EMP)

Change Information No change issue detected.

Links

Artifact [top_level_arch.slx](#)

 [Mass/Surge Production & Storage](#) (↩Implemented by)

Artifact [Avionics tests.m](#)

 [function IgnitionSystem\(testCase\)](#) (↩Verified by)

Implementation Status

Total	Implemented	Justified	None
1	1	0	0

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	1	0	0	0	0

33 Software

Requirement Type Functional

ID RTCA-DO-178

Description

Missile software will comply to military software guidance

Change Information No change issue detected.

Links

Artifact [top_level_arch.slx](#)

 [Software](#) (←Implemented by)

Artifact [Avionics tests.m](#)

 [function Software\(testCase\)](#) (←Verified by)

Implementation Status

Total	Implemented	Justified	None
1	1	0	0

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	1	0	0	0	0

34 Connection Interface

Requirement Type Functional

ID MIL-STD-1760

Description

Missile will be compatible to standard aircraft weapon connection interface

Change Information No change issue detected.

Links

Artifact [top_level_arch.slx](#)

 [Software](#) (←Implemented by)

Artifact [Avionics tests.m](#)

 [function Connection_standard\(testCase\)](#) (←Verified by)

Implementation Status

Total	Implemented	Justified	None
1	1	0	0

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	1	0	0	0	0

35 Electric Power Standard

Requirement Type Functional

ID MIL-STD-704

Description

Missile will comply to military electric power standard

Change Information No change issue detected.

Links

Artifact [top_level_arch.slx](#)

 [Battery](#) (←Implemented by)

Artifact [Avionics tests.m](#)

 [function Power Standard\(testCase\)](#) (↩Verified by)

Implementation Status

Total	Implemented	Justified	None
1	1	0	0

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	1	0	0	0	0

36 Data Link System

Requirement Type Functional

ID MIL-STD-6016

Description

Missile will be compatible to link 16 data link system

Change Information No change issue detected.

Links

Artifact [top_level_arch.slx](#)

☐ [Avionics Package](#) (↩Implemented by)

Artifact [Avionics tests.m](#)

 [function Datalink\(testCase\)](#) (↩Verified by)

Implementation Status

Total	Implemented	Justified	None
1	1	0	0

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	1	0	0	0	0

37 Electromagnetic Resistance

Requirement Type Functional

ID MIL-STD-461

Description

Missile will be redundant to electromagnetic interference

Change Information No change issue detected.

Links

Artifact [top_level_arch.slx](#)

☐ [Available Components](#) (↩Implemented by)

Artifact [Avionics tests.m](#)

 [function Electromagnetic Resistance\(testCase\)](#) (↩Verified by)

Implementation Status

Total	Implemented	Justified	None
1	1	0	0

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	1	0	0	0	0

Appendix

Artifact List

Simulink models:

#	Name	Folder	Version
1	top_level_arch.slx	C:\Users\andgo\Documents\GitHub\AVD\AOE4066_Team1_Scorpius\Platform	1.37

Others

#	Name	Folder	File timestamp
1	UnitCost_Test.m	C:\Users\andgo\Documents\GitHub\AVD\AOE4066_Team1_Scorpius\Tests	03-May-2026 18:14:47
2	Production_Tests.m	C:\Users\andgo\Documents\GitHub\AVD\AOE4066_Team1_Scorpius\Tests	03-May-2026 18:14:18
3	Cruise_Metrics_Test.m	C:\Users\andgo\Documents\GitHub\AVD\AOE4066_Team1_Scorpius\Tests	03-May-2026 18:10:59
4	Weight_Test.s.m	C:\Users\andgo\Documents\GitHub\AVD\AOE4066_Team1_Scorpius\Tests	03-May-2026 18:04:03
5	Warhead_Size_Destruction.m	C:\Users\andgo\Documents\GitHub\AVD\AOE4066_Team1_Scorpius\Tests	03-May-2026 18:05:08
6	hit_the_target.m	C:\Users\andgo\Documents\GitHub\AVD\AOE4066_Team1_Scorpius\Tests	03-May-2026 18:11:53
7	LaunchBox_Size_Test.m	C:\Users\andgo\Documents\GitHub\AVD\AOE4066_Team1_Scorpius\Tests	03-May-2026 18:12:11
8	Avionics_test.s.m	C:\Users\andgo\Documents\GitHub\AVD\AOE4066_Team1_Scorpius\Tests	03-May-2026 18:10:21
9	Lowlevel_ingress_Test.m	C:\Users\andgo\Documents\GitHub\AVD\AOE4066_Team1_Scorpius\Tests	03-May-2026 18:05:26
10	Material_Tests.m	C:\Users\andgo\Documents\GitHub\AVD\AOE4066_Team1_Scorpius\Tests	03-May-2026 18:12:50